# Resolution Geometry: Quantum Gravity

## A Conditional Derivation of the Quantum State Structure, Causal Localization, and Gravitational Effective Field Theory

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**Abstract**

A derivation of low-energy quantum gravity is developed from operational records under an explicit hierarchy of state-space, composition, causal-realization, and metric-response assumptions. Operational equivalence first determines which properties and updates descend to the physical state description. Polytope rigidity and an all-dimension composition identity isolate necessary features of a nonclassical locally tomographic theory; a stated reconstruction theorem supplies the complex quantum realization. Positive distinguishability cannot itself produce Lorentzian signature. Causal order and calibrated volume instead determine a metric within an admitted distinguishing Lorentzian realization. A finite-range unitary quantum walk supplies an exact microscopic propagation cone, a discrete operator continuity equation, graded locality, and a controlled free Dirac limit. These constructions establish causal localization, not dynamical gravity by themselves.

A covariant source functional then identifies the stress response and its Ward identity. Within a local two-derivative symmetric-tensor sector, gauge invariance selects the linearized Einstein operator. With the stated nonlinear consistency assumptions, the leading action is Einstein–Hilbert with universal source coupling. Perturbative quantization yields gravitational effective field theory rather than Einstein-only quantum closure. Relational observables, stress fluctuations, and a branch-phase entanglement prediction connect the construction to measurable records. Quantitative distinguishability bounds delimit finite-resource inference without identifying small effects with exact observational nullity. The theorem ladder distinguishes results proved here, structural results used from the literature, conditional physical identifications, and unresolved nonperturbative questions. No unique ultraviolet completion, interacting four-dimensional microscopic continuum limit, or anomaly-free quantum constraint representation is asserted.

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# 1 Operational foundations and the derivation statement

The object to be derived is a low-energy quantum theory of a dynamical metric coupled to matter, with a controlled relation between physical records, gauge equivalence, causal localization, and observables. A classical metric equation and a quantum state space are not sufficient separately: the source response, composition, and approximation connecting them must also be specified.

The argument is organized by logical dependence rather than by the chronology of a construction. Mathematical statements with proofs are distinguished from structural inputs whose hypotheses are retained. A finite numerical calculation witnesses an identity or an example; it does not establish a reconstruction theorem, a continuum existence theorem, or ultraviolet completeness.

**Definition 1.1** (Operational equivalence)**.** Let $X$ be a candidate state space and let $P_x^a$ be the outcome law of a declared protocol $a \in \mathcal{P}$. Define

$$x \sim_{\mathcal{P}} y \quad \Longleftrightarrow \quad P_x^a = P_y^a \quad \text{for every } a \in \mathcal{P}, \qquad Q = X/\!\sim_{\mathcal{P}}. \tag{1}$$

Equality here is exact equality of laws, not failure of a finite experiment to reject a hypothesis.

**Theorem 1.2** (Descent of properties and dynamics)**.** *A property $C : X \to Y$ is a property of $Q$ if and only if it is constant on every equivalence class. A deterministic update $U : X \to X$ induces an update on $Q$ if and only if*

$$x \sim_{\mathcal{P}} y \implies Ux \sim_{\mathcal{P}} Uy. \tag{2}$$

*Proof.* If $C = \bar{C} \circ \pi$ for the quotient map $\pi$, equal quotient points have equal $C$. Conversely, constancy makes $\bar{C}([x]) = C(x)$ well defined. The same argument applied to $\bar{U}([x]) = [Ux]$ proves the dynamical statement. □

For stochastic dynamics, the corresponding condition is equality of pushed-forward transition laws on the quotient. In a finite Markov chain it is strong lumpability: the transition probability into each quotient block is independent of the representative within the starting block. Present agreement alone does not imply this condition.

**Assumption A1** (Operational closure)**.** The retained records separate the states relevant to the intended predictions and satisfy predictive descent. If a future experiment distinguishes representatives, the state description is refined rather than assigned an autonomous update prematurely.

**Scope of the endpoint.** The principal result, Theorem 17.1, is conditional. Its assumptions identify a quantum realization, an admissible Lorentzian continuum, and a universal healthy tensor response. Its conclusion is the gravitational effective field theory (EFT) and its observable low-energy consequences. A microscopic derivation of all these assumptions is a separate existence and selection question.

## 2 Quantum state structure and composition

**Assumption A2** (Finite-dimensional operational systems)**.** Each elementary system under consideration has a finite-dimensional compact convex normalized state space and finite information capacity. Effects are affine outcome-probability functionals; composites and their admissible effects are part of the theory.

Finite capacity and finite dimension are separate requirements. A bound on the number of perfectly distinguishable states does not, without additional hypotheses, imply a finite-dimensional affine state space. Likewise, a finite experimental resolution does not turn every continuum classical model into a simplex. Entropy bounds can motivate finite-capacity descriptions in their domain of applicability, but do not replace these operational assumptions [1].

**Assumption A3** (Continuous reversible transitivity)**.** For each elementary closed system, the connected reversible transformation group acts transitively on pure states. The requirement concerns closed-system transformations, not arbitrary reduced dynamics after discarding an environment.

**Assumption A4** (Composition and reconstruction conditions)**.** Composite states are determined by joint probabilities of product effects (local tomography). The theory also satisfies the subspace-equivalence and elementary-system measurement conditions of the quantum reconstruction used below. In particular, all effects required by that reconstruction on the elementary bit are allowed. These are substantive assumptions about a family of systems, not consequences of one interference experiment.

**Theorem 2.1** (Polytope rigidity)**.** *The affine automorphism group of a finite-dimensional polytope is finite. Every continuous one-parameter group of its affine automorphisms is trivial.*

*Proof.* Every affine automorphism permutes the finite vertex set. The permutation action is faithful because the vertices affinely span the polytope. The automorphism group therefore embeds in a finite permutation group. A continuous map from a connected parameter interval to that finite group is constant. □

**Corollary 2.2** (Exclusion of nontrivial polytopic pure-state motion)**.** *A polytope with more than one pure state cannot satisfy Assumption A3. This excludes finite classical simplices and finite polygon models within the declared class. It does not exclude a classical continuum with unrestricted effects: such a model can have continuous pure-state motion and unbounded capacity.*

**Theorem 2.3** (All-dimension matrix-family composition identity)**.** *Within a same-type matrix family with rank $d$, parameter count*

$$K_k(d) = d + \frac{k}{2}d(d-1), \tag{3}$$

*and composite rank $d_{AB} = d_A d_B$, local tomography requires*

$$K_k(ab) - K_k(a)K_k(b) = -\frac{k(k-2)}{4}ab(a-1)(b-1) = 0. \tag{4}$$

*For $a, b > 1$, the only possibilities are $k = 0$ and $k = 2$.*

*Proof.* Substitution and factorization give (4). The remaining factors are nonzero for nontrivial ranks. □

The identity is exact in all ranks, not a finite-rank extrapolation. The classical count $k = 0$ and the complex matrix count $k = 2$ survive; the real and quaternionic counts $k = 1, 4$ do not satisfy this particular composition rule. The calculation does not classify arbitrary convex theories, arbitrary spin factors, or exceptional Jordan systems. In particular, the exceptional octonionic algebra is not an unrestricted all-rank matrix family.

**Structural input 2.4** (Finite-dimensional quantum reconstruction)**.** *The reconstruction of Masanes and Müller uses finite-dimensional operational systems, continuous reversible pure-state transitivity, tomographic locality, equivalence of subspaces, and its stated elementary-system measurement assumption to obtain finite-dimensional complex quantum theory [2]. Assumptions A2–A4 are understood with the full operational content of that theorem, not merely the parameter-count identity. Other reconstructions use different sufficient axiom sets [3, 4].*

Consequently the quantum representation used here is

$$\rho \succeq 0, \quad \operatorname{Tr}\rho = 1, \qquad 0 \preceq E \preceq I, \qquad P(E \mid \rho) = \operatorname{Tr}(\rho E), \qquad \mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B. \tag{5}$$

This is a conditional identification by a reconstruction theorem. Polytope rigidity and composition counting explain two restrictions but do not substitute for the full reconstruction. The state-space axioms do not determine a gravitational Hamiltonian or select the matter spectrum.

## 3 Coherence, reference structure, and distinguishability

Fix a reference basis and define dephasing by $\Delta(\rho) = \operatorname{diag}\rho$. Diagonal unitary rephasing acts as $\rho \mapsto D\rho D^\dagger$. The two constructions have different quotients.

**Proposition 3.1** (Dephasing fibres and phase orbits)**.** *Dephasing is the uniform diagonal-phase twirl, but equality of dephased states does not imply membership in the same diagonal-unitary orbit.*

*Proof.* Integrating $e^{i(\theta_i - \theta_j)}\rho_{ij}$ over independent phases kills every off-diagonal entry. However, diagonal rephasing preserves $|\rho_{ij}|$ and the spectrum. The states $(I + aX)/2$ with different $|a|$ all dephase to $I/2$ and have different spectra. They therefore lie in different orbits. □

On a fixed nonzero support graph, diagonal rephasing changes an edge phase by $\theta_i - \theta_j$. A spanning forest removes all tree-edge phases. The remaining independent cycle phases number $|E| - |V| + c$, where $c$ is the number of connected components. Products such as $\rho_{01}\rho_{12}\rho_{20}$ are invariant. Positivity restricts the allowed edge magnitudes and phases; support changes can alter the orbit stratum.

**Proposition 3.2** (Operational recovery of off-diagonal structure)**.** *Let $p_i = \rho_{ii}$, and let $p^+_{ij}$ and $p^i_{ij}$ be probabilities of the rank-one effects associated with $(|i\rangle + |j\rangle)/\sqrt{2}$ and $(|i\rangle + i|j\rangle)/\sqrt{2}$. Then*

$$\operatorname{Re}\rho_{ij} = p^+_{ij} - \frac{p_i + p_j}{2}, \qquad \operatorname{Im}\rho_{ij} = \frac{p_i + p_j}{2} - p^i_{ij}. \tag{6}$$

*Thus populations and two quadratures for every pair identify the density operator in the declared finite-dimensional realization.*

*Proof.* Expand the two Born probabilities. Hermiticity fixes the conjugate entry. The resulting entries exhaust a Hermitian matrix, subject to trace normalization and positivity. □

Uncertainty in a fixed measurement is not coherence: an eigenstate can have zero uncertainty and zero basis coherence, while a coherent superposition has nonzero uncertainty in that basis. Conversely, a scalar coherence measure does not identify the state or its cycle phases. These distinctions prevent a scalar information density from being used as a complete gravitational source.

On a regular fixed-rank stratum and for tangent directions with finite information, the symmetric logarithmic derivative $L$ satisfies $\dot\rho = (\rho L + L\rho)/2$ on the relevant support. The SLD quadratic form is $F_Q = \operatorname{Tr}(\rho L^2)$, with Bures convention $\mathrm{d}s^2_B = F_Q \mathrm{d}\theta^2/4$ for a one-parameter family. Directional optimization over measurements does not guarantee one measurement jointly attains the full multiparameter matrix.

**Proposition 3.3** (A joint single-copy resolution obstruction)**.** *At $\rho = I/2$ with tangent derivatives $X/2, Y/2$, the SLD Fisher matrix is $I_2$. Every single-copy qubit POVM has classical Fisher matrix satisfying $\operatorname{Tr} F_{xy} \le 1$, so cannot attain $I_2$.*

*Proof.* Write each nonzero effect as $E_a = w_a(I + n_a \cdot \sigma)$, where $|n_a| \le 1$, $\sum_a w_a = 1$, and $\sum_a w_a n_a = 0$. At the center $p_a = w_a$ and $\partial_i p_a = w_a n_{a,i}$. Hence $F_{xy} = \sum_a w_a n_{a,xy} n^T_{a,xy}$ and $\operatorname{Tr} F_{xy} \le \sum_a w_a = 1$. □

The statement is single-copy and local in parameter space. It does not rule out a different collective asymptotic estimation task. Likewise, the Berry curvature of the full pure-state manifold must not be confused with its pullback to a fixed-population fibre. For a pure qubit that fibre is a circle, on which every two-form vanishes; the Chern number belongs to the full projective-state sphere.

# 4 Causal realization and metric scale

**Obstruction 4.1** (Signature cannot arise by a real distinguishability pullback)**.** *If $G \succeq 0$ and $J$ is real, then $J^T G J \succeq 0$. No real pullback of a positive Fisher, SLD, or Bures metric has Lorentzian signature.*

*Proof.* For every $v$, $v^T J^T G J v = (Jv)^T G (Jv) \ge 0$. □

A Lorentzian realization therefore requires causal or dynamical structure beyond a positive state-space metric. Analytic continuation or a subtraction rule is additional structure, not a coordinate change proving the desired signature.

**Assumption A5** (Admissible causal continuum)**.** The relevant records admit a smooth, time-oriented, distinguishing Lorentzian realization of dimension $d \ge 3$, with a specified correspondence between operational reachability and the causal or chronological relation used in the reconstruction theorem. A calibrated local volume measure is also supplied. Dimension, manifoldlikeness, and this correspondence are not inferred from an arbitrary partial order alone.

**Structural input 4.2** (Causal reconstruction)**.** *Under the distinguishing and regularity hypotheses of the causal reconstruction results, preservation of the appropriate causal/chronological structure determines the Lorentzian conformal geometry [6, 7]. The conclusion applies within an admitted Lorentzian manifold class; it is not an existence theorem assigning a manifold to every ordered set.*

**Theorem 4.3** (Calibrated volume fixes the conformal factor)**.** *Suppose two smooth Lorentzian metrics on the same d-manifold have the same time-oriented causal structure under the reconstruction hypotheses and the same local volume measure. Then they coincide under the resulting conformal identification.*

*Proof.* Causal reconstruction gives $\widetilde{g} = \Omega^2 g$ with $\Omega > 0$. Their volume forms satisfy $\mathrm{d}V_{\widetilde{g}} = \Omega^d \mathrm{d}V_g$. Equality of measures on all relatively compact regions implies $\Omega^d = 1$ almost everywhere, and smoothness gives $\Omega = 1$ everywhere. □

**Proposition 4.4** (Statistical volume calibration)**.** *For a Poisson sprinkling of known density $\varrho$ into a region of volume $V$, $\widehat{V} = N/\varrho$ is unbiased and*

$$\operatorname{Var} \widehat{V} = V/\varrho, \qquad \frac{\operatorname{sd} \widehat{V}}{V} = \frac{1}{\sqrt{\varrho V}}. \tag{7}$$

*Proof.* The count has mean and variance $\varrho V$. □

Counting supplies a statistical estimate of volume once density and units are calibrated [8]. If $\varrho$ is unknown, the degeneracy $\varrho \mapsto \varrho/\Omega^d$, $g \mapsto \Omega^2 g$ remains. A finite count is not exact geometry. The physical dimension and an interacting continuum limit remain separate obligations.

## 5 A finite-range unitary realization and its free continuum limit

A local reversible construction establishes compatibility of exact microscopic propagation bounds with a relativistic free limit. It is an existence example, not a unique microscopic law or a four-dimensional gravitational model.

On $\ell^2(\mathbb{Z}) \otimes \mathbb{C}^2$, define

$$C_m = \begin{pmatrix} n & -im \\ -im & n \end{pmatrix}, \quad n = \sqrt{1-m^2}, \quad 0 \le m < 1, \qquad U = SC_m, \tag{8}$$

where $S$ shifts the right component one site right and the left component one site left. This is a coined quantum walk with a Dirac-type low-energy limit. Its precise update is fixed here; distinct Dirac automata can have different finite-spacing dispersion. A low-momentum sector does not remove other Brillouin-zone branches; species multiplicity and fermion doubling require separate analysis in a microscopic matter model.

**Theorem 5.1** (Exact finite propagation and dispersion)**.** *The update is unitary and has propagation range one. A state initially supported on $A$ is supported on its $t$-neighborhood after $t$ integer steps. In Fourier space,*

$$U(k) = e^{-ikZ} C_m, \qquad \cos\omega(k) = n \cos k. \tag{9}$$

*For $0 < m < 1$,*

$$\left|\frac{\mathrm{d}\omega}{\mathrm{d}k}\right| \le 1, \qquad \omega(k)^2 = \alpha^2 + \frac{\alpha n}{m} k^2 + O(k^4), \quad \alpha = \arcsin m. \tag{10}$$

*For $m = 0$, the positive quasienergy branch is $|k|$ on the principal Brillouin interval.*

*Proof.* The coin and conditional shift are unitary. Each step enlarges support by at most one site, so induction proves exact finite propagation. The Fourier matrix has determinant one and trace $2n\cos k$, giving the dispersion. Differentiation gives

$$\omega'(k)^2 = \frac{n^2 \sin^2 k}{1 - n^2 \cos^2 k} \leq 1, \tag{11}$$

because the denominator minus the numerator is $m^2$. Expanding $\omega^2$ at $k = 0$ gives (10). □

In particular,

$$\alpha^2 = m^2 + \frac{m^4}{3} + O(m^6), \qquad \frac{\alpha n}{m} = 1 - \frac{m^2}{3} - \frac{2m^4}{15} + O(m^6). \tag{12}$$

These are expansions, not exact polynomial dispersion relations. Their even powers do not by themselves certify all observational Lorentz tests for an interacting higher-dimensional model.

**Theorem 5.2** (Controlled free Dirac limit on a momentum band)**.** *Let the spatial and temporal spacing be $a$, set $m = \sin(Ma)$ for fixed $M \geq 0$, and restrict to $|p| \leq P$. For $N = T/a \in \mathbb{N}$ and sufficiently small $a$,*

$$U_a(p) = e^{-iapZ} e^{-iaMX}, \qquad H_D(p) = pZ + MX, \tag{13}$$

*satisfy*

$$\sup_{|p| \leq P} \left\| U_a(p)^N - e^{-iTH_D(p)} \right\| \leq TaP|M|. \tag{14}$$

*Proof.* For bounded self-adjoint $A, B$, the Duhamel product estimate gives $\|e^{-iaA}e^{-iaB} - e^{-ia(A+B)}\| \leq a^2\|[A,B]\|/2$. With $A = pZ$ and $B = MX$, the right side is $a^2|pM|$. A telescoping sum of $N$ products of unitaries gives $Na^2|pM| \leq TaP|M|$. Fourier transformation transfers the uniform multiplier bound to the band-limited Hilbert space. □

This closes a free, finite-time, fixed-band convergence statement. It does not control arbitrarily large momentum, interacting renormalization, or a dynamical curved geometry. The state-evolution error in (14) is $O(a)$; an $O(a^2)$ quasienergy correction is a different assertion. Weyl and Dirac automata in higher dimensions provide related constructions under their own graph, isotropy, and representation hypotheses [10, 11].

## 6 Local observable algebras and exact quantum balance

Let $a_{x,s}$ satisfy the canonical anticommutation relations (CAR). A one-particle unitary induces a CAR automorphism, with field evolution $a(t) = U^t a(0)$ in a consistent convention.

**Theorem 6.1** (Graded locality and observable locality)**.** *For the finite-range update,*

$$\{a_{x,s}(t), a^\dagger_{y,r}(0)\} = (U^t)_{xs,yr} = 0 \quad \text{if } |x - y| > t. \tag{15}$$

*Odd fields have graded locality. Even observable algebras supported in disjoint causal neighborhoods commute.*

*Proof.* The CAR reduce the unequal-time anticommutator to the one-particle propagator, whose support is bounded by Theorem 5.1. Products of even parity in disjoint CAR supports commute because moving all generators past one another produces an even number of sign changes. □

The even local net is the observable-locality object. The odd field net is graded-local, not an ordinary commuting net. A lattice net with this property is not yet a continuum Haag–Kastler theory with all covariance, spectrum, and representation axioms [9].

**Theorem 6.2** (Operator continuity equation)**.** *Let $b_x = C_m a_x$ be the post-coin fields, $n_x = a^\dagger_{x,R} a_{x,R} + a^\dagger_{x,L} a_{x,L}$, and*

$$J_{x+1/2} = b^\dagger_{x,R} b_{x,R} - b^\dagger_{x+1,L} b_{x+1,L}. \tag{16}$$

*Then*

$$n_x(t+1) - n_x(t) = -\big(J_{x+1/2}(t) - J_{x-1/2}(t)\big) \tag{17}$$

*as an operator identity.*

*Proof.* Coin unitarity gives $n_x = b^\dagger_{x,R} b_{x,R} + b^\dagger_{x,L} b_{x,L}$. After the shift, $n_x(t+1) = b^\dagger_{x-1,R} b_{x-1,R} + b^\dagger_{x+1,L} b_{x+1,L}$. Subtraction is exactly (17). □

The identity holds in every state, not only for one simulated wave packet. It is a number-current balance law. Identifying it with energy-momentum conservation requires an additional stress construction; a conserved probability current alone does not identify the gravitational source.

## 7 Covariant localization and the stress response

A local net assigns observable algebras to already specified regions. It establishes localization and inclusion relations. It does not, by itself, determine a metric, a dynamical tetrad, or the map from microscopic records to a gravitational response.

**Assumption A6** (Covariant source realization)**.** In the admitted continuum regime, local matter records admit a covariant renormalized generating functional $W[g, J]$ on admissible backgrounds. The metric argument is the physical metric calibrated by causal and volume records. Its variations exist to the required order, obey the appropriate diffeomorphism Ward identity, and are compatible with the retained microscopic records. Scalar sources $J$ are used below for simplicity. Boundary conditions and additional source transformations are specified when needed.

This assumption is the state-to-spacetime identification needed for a stress response. It is stronger than merely possessing a local net, and is not supplied by an internal phase holonomy.

**Theorem 7.1** (Stress response and Ward identity)**.** *Define the matter stress by*

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}, \tag{18}$$

*and the renormalized expectation and scalar response by*

$$\delta W = \int \sqrt{-g} \left( \frac{1}{2} \langle T^{\mu\nu} \rangle \delta g_{\mu\nu} + \langle \mathcal{O} \rangle \delta J \right). \tag{19}$$

*For compactly supported infinitesimal diffeomorphisms and no diffeomorphism anomaly,*

$$\nabla_\mu \langle T^\mu{}_\nu \rangle = \langle \mathcal{O} \rangle \nabla_\nu J. \tag{20}$$

*Functional differentiation gives the corresponding stress-insertion Ward identities with contact terms.*

*Proof.* Use $\delta_\xi g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu$ and $\delta_\xi J = \xi^\nu \nabla_\nu J$. Integrating the metric term by parts in $\delta_\xi W = 0$ gives (20) because $\xi$ is arbitrary. Further source derivatives differentiate both the expectation values and their transformation laws, producing the contact terms. □

Conservation does not fix all source couplings. In flat space,

$$I_{\mu\nu} = (\eta_{\mu\nu}\Box - \partial_\mu\partial_\nu)F, \qquad \partial^\mu I_{\mu\nu} = 0, \tag{21}$$

is an improvement, related to a nonminimal $RF$ coupling in a covariant action. Scalar improvements, spin/torsion-sensitive sectors, and boundary responses must therefore be retained in the source ledger.

**Obstruction 7.2** (Internal phase curvature does not determine gravity)**.** *A nonzero cycle phase of a finite-dimensional quantum state does not imply nonzero spacetime Riemann curvature.*

*Proof.* Take a positive three-level density matrix with nonzero $\arg(\rho_{01}\rho_{12}\rho_{20})$ and realize it as an internal state on a fixed Minkowski background. The internal invariant is nonzero while the background Riemann tensor vanishes. □

The Coleman–Mandula theorem concerns symmetry algebras of an analytic scattering theory under specific hypotheses, including a mass gap [13]. It is not a general proof that a microscopic internal structure can never participate in an emergent spacetime construction, and it is not directly applicable to a massless gravitational theory without checking those hypotheses. The obstruction above is narrower and sufficient: internal holonomy alone does not perform the required identification.

## 8 Controlled limits and source fluctuations

A continuum limit must control the source and its variations, not only the underlying field values.

**Obstruction 8.1** (Nonlinear stress need not commute with a field limit)**.** *Uniform convergence of fields does not imply convergence of their quadratic gradient stresses to the stress of the limiting field.*

*Proof.* On a periodic interval, $u_n(x) = \sin(nx)/n$ converges uniformly to zero, but

$$\frac{1}{2}|\partial_x u_n|^2 = \frac{1}{2}\cos^2(nx) \rightharpoonup \frac{1}{4}, \tag{22}$$

whereas the limiting field has zero gradient stress. □

A strong enough convergence topology, suitable compactness, or an explicit defect stress is required. At the quantum level, products and stress correlators additionally require renormalization. A finite-range free limit does not settle these interacting source limits.

The mean-source equation

$$G_{\mu\nu}[g] + \Lambda g_{\mu\nu} = 8\pi G\langle T_{\mu\nu}\rangle_{\mathrm{ren}} \tag{23}$$

is a semiclassical approximation, not the definition of the full quantum theory. Equal means need not give equal fluctuations. For $H = \mathrm{diag}(0, 1, 2)$, the states $\mathrm{diag}(0, 1, 0)$ and $\mathrm{diag}(1/2, 0, 1/2)$ both have mean energy one but variances zero and one. A quantum source description retains connected stress correlations and their backreaction in the regime where they matter.

## 9 The universal tensor response

**Assumption A7** (Leading gravitational sector)**.** In a four-dimensional Lorentzian regime, the long-range gravitational response is carried by one healthy massless symmetric tensor. Its leading local parity-even kinetic operator has two derivatives, the matter response is that of Assumption A6, and gauge relabelings preserve physical coincidences. Additional fields and nonminimal couplings are declared explicitly. The existence of this sector and the physical dimension are assumptions, not consequences of positive distinguishability alone.

**Theorem 9.1** (Linear tensor selection). *Let $v_\nu = \partial^\alpha h_{\alpha\nu}$ and $h = \eta^{\mu\nu} h_{\mu\nu}$. In the class*

$$\begin{aligned}\mathcal{E}_{\mu\nu}(h) = A\Box h_{\mu\nu} + B(\partial_\mu v_\nu + \partial_\nu v_\mu) + C\partial_\mu\partial_\nu h \\ + \eta_{\mu\nu}(D\partial\cdot v + E\Box h),\end{aligned} \tag{24}$$

*gauge invariance under $\delta h_{\mu\nu} = \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$ and the identity $\partial^\mu\mathcal{E}_{\mu\nu} = 0$ select*

$$(A, B, C, D, E) = A(1, -1, 1, 1, -1). \tag{25}$$

*The resulting operator is proportional to the linearized Einstein tensor.*

*Proof.* The divergence has independent coefficients $A + B$, $B + D$, and $C + E$, so $B = -A$, $D = A$, and $E = -C$. On a pure gauge field, the coefficients of $\partial_\mu\partial_\nu\partial\cdot\xi$ and $\eta_{\mu\nu}\Box\partial\cdot\xi$ are respectively $2(B + C)$ and $2(D + E)$. Their vanishing gives $C = A$, completing the selection. Substitution verifies both identities for arbitrary $h$ and $\xi$. □

The overall healthy sign and normalization remain to be fixed. In the transverse-traceless wave sector in four dimensions, the spatial symmetric tensor has six components; transversality to a fixed wave vector and tracelessness leave two independent polarizations. This is a linear physical-sector count, not a proof of nonlinear constraint closure.

**Proposition 9.2** (Conserved-source gauge decoupling). *The linear coupling $\frac{\kappa}{2}\int h_{\mu\nu}T^{\mu\nu}$ is invariant under compactly supported linearized gauge transformations when $\partial_\mu T^{\mu\nu} = 0$.*

*Proof.* Its variation is $\kappa\int\partial_\mu\xi_\nu T^{\mu\nu} = -\kappa\int\xi_\nu\partial_\mu T^{\mu\nu}$. □

**Structural input 9.3** (Universal coupling and nonlinear completion). *For a local healthy massless spin-two field with the declared field content and derivative order, consistent self-coupling and diffeomorphism invariance give the Einstein leading sector, up to the cosmological term, boundary terms, field redefinitions, and declared nonminimal matter couplings. A canonical uniqueness route uses the hypersurface-deformation algebra with its locality, phase-space, and regularity hypotheses [19]. The full nonlinear consistency result is used as structural input; the linear proof above is not presented as its replacement.*

In an admissible Lorentz-invariant factorizing scattering regime, the leading soft spin-two factor has the form

$$S_{\text{soft}} \propto \sum_i \eta_i g_i \frac{p_i^\mu p_i^\nu \epsilon_{\mu\nu}}{p_i\cdot q}. \tag{26}$$

Gauge decoupling requires $\sum_i \eta_i g_i p_i^\nu = 0$. For generic connected processes, comparison with momentum conservation requires common coupling. This is a soft-factor argument under scattering assumptions, not a global scattering construction for arbitrary cosmological backgrounds.

**Corollary 9.4** (Classical leading equation). *Under Assumptions A6 and A7 and Structural input 9.3, choose*

$$S_{\text{EH}} = \frac{2}{\kappa^2}\int\sqrt{-g}(R - 2\Lambda), \qquad \kappa^2 = 32\pi G. \tag{27}$$

*Stationarity of $S_{\text{EH}} + S_m$, with the required boundary treatment, gives*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{28}$$

*Proof.* Variation with respect to $g^{\mu\nu}$ gives $2(G_{\mu\nu}+\Lambda g_{\mu\nu})/\kappa^2 - T_{\mu\nu}/2 = 0$. The stated normalization follows. □

The Bianchi identity supplies compatibility with covariant conservation. It does not determine the numerical values of $G$, $\Lambda$, or the nonminimal couplings.

## 10 Perturbative quantization and anomaly conditions

**Assumption A8** (Perturbative quantum consistency)**.** The admitted quantum theory is quantized around a suitable background with a controlled low-energy expansion, gauge fixing, ghosts, renormalization conditions, and no uncanceled gauge or diffeomorphism anomaly in the specified matter content. Predictions are defined through gauge-independent observables with an appropriate infrared prescription.

Write $g = \bar{g} + \kappa h$. The formal generating functional is

$$Z[J] = \int \mathcal{D}h\, \mathcal{D}\psi\, \mathcal{D}\bar{c}\, \mathcal{D}c\; e^{i(S_{\mathrm{EFT}} + S_{\mathrm{gf}} + S_{\mathrm{gh}} + J\cdot\mathcal{O})}. \tag{29}$$

This denotes a perturbative construction. It is not a convergent nonperturbative Lorentzian measure. BRST/Ward identities, physical-state positivity, and regulator dependence must be controlled at the order used.

**Proposition 10.1** (Anomaly controls for one Standard-Model generation)**.** *For left-handed Weyl multiplets $Q, u^c, d^c, L, e^c$ with hypercharges $1/6, -2/3, 1/3, -1/2, 1$, respectively, the $U(1)_Y$ mixed gauge–gravitational, cubic hypercharge, $SU(2)^2U(1)$, and $SU(3)^2U(1)$ anomaly sums vanish. The $SU(3)^3$ chiral anomaly cancels, and the number of $SU(2)$ doublets is even.*

*Proof.* Including color and weak multiplicities,

$$\operatorname{Tr} Y = 6/6 - 3(2/3) + 3/3 - 2/2 + 1 = 0, \tag{30}$$

$$\operatorname{Tr} Y^3 = 6(1/6)^3 + 3(-2/3)^3 + 3(1/3)^3 + 2(-1/2)^3 + 1 = 0. \tag{31}$$

Up to common Dynkin-index factors, the mixed sums are $3(1/6) - 1/2 = 0$ and $2(1/6) - 2/3 + 1/3 = 0$. The cubic color count is $2 - 1 - 1 = 0$, and the weak-doublet count is $3 + 1 = 4$. Removing $e^c$ makes both hypercharge traces equal to $-1$. □

These are specified matter-sector checks. The first is a mixed $U(1)$–gravity anomaly, not a proof of all gravitational consistency. The familiar four-dimensional local pure diffeomorphism anomaly is not the same anomaly polynomial. Global gauge-group choices, boundaries, additional matter, and nonperturbative gravitational constraints require their own treatment. The matter spectrum is an empirical input, not selected by the cancellation arithmetic alone.

## 11 The gravitational effective action

**Proposition 11.1** (Derivative power counting)**.** *For a connected graph in four dimensions with $L$ loops and vertices carrying $d_v$ derivatives, the superficial momentum degree is*

$$D = 2 + 2L + \sum_v (d_v - 2). \tag{32}$$

*Proof.* With $I$ internal lines and $V$ vertices, $D = 4L - 2I + \sum_v d_v$. Use $I = L + V - 1$. □

Power counting identifies allowed counterterms, not their nonzero coefficients. For pure Einstein gravity with vanishing cosmological constant, the one-loop on-shell ultraviolet divergence cancels modulo the four-dimensional Euler density and field redefinitions [14]. This is not a statement that every one-loop amplitude is finite: infrared divergences, matter, boundaries, and different backgrounds require separate analysis.

**Structural input 11.2** (Two-loop Einstein nonclosure)**.** *Pure four-dimensional Einstein gravity has a nonzero two-loop on-shell ultraviolet counterterm cubic in the Riemann tensor [15]. Its precise coefficient depends on action and regulator conventions; the nonzero result is the structural fact used here.*

For the curvature convention of the accompanying symbolic calculation, Schwarzschild gives

$$R = 0, \quad R_{\mu\nu}R^{\mu\nu} = 0, \quad R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{48M^2}{r^6}, \quad R^{ab}{}_{cd}R^{cd}{}_{ef}R^{ef}{}_{ab} = \frac{96M^3}{r^9}. \tag{33}$$

The cubic invariant is nonzero on a Ricci-flat background. Its value is a witness, not by itself a proof of a loop coefficient or a complete cohomological nonremovability classification; that conclusion uses Structural input 11.2.

**Theorem 11.3** (Conditional low-energy quantum gravitational action)**.** *Under Assumption A8, the symmetry-compatible low-energy action is an EFT of the form*

$$S_{\mathrm{EFT}} = \int \sqrt{-g}\left[\frac{M_{\mathrm{Pl}}^2}{2}(R - 2\Lambda) + \mathcal{L}_m + \sum_i c_i \mathcal{O}_i^{(4)} + \sum_j \frac{d_j}{M^2}\mathcal{O}_j^{(6)} + \cdots\right], \qquad M_{\mathrm{Pl}}^{-2} = 8\pi G. \tag{34}$$

*For a fixed field content, specified observable, and fixed EFT power-counting order, finitely many independent coefficients enter. The power counting includes canonical dimension or a specified field-amplitude expansion as well as derivatives and loops; derivative order alone need not bound arbitrary scalar-field functions. Einstein-only perturbative closure is excluded by Structural input 11.2; low-energy predictivity is not excluded [16].*

*Proof.* Local counterterms and matching terms are organized by the symmetries and derivative order. At a fixed order in that power counting the independent operator basis is finite after identities, integrations by parts, and field redefinitions. Renormalization fixes scale dependence; matching or measurements fix the required finite coefficients. The nonzero cubic counterterm cannot be absorbed by renormalizing only $G$ and $\Lambda$. □

In four dimensions the curvature-squared coefficients are dimensionless. The Euler density, total derivatives, and field redefinitions reduce the independent basis; with matter, the induced contact and nonminimal terms must be retained. Light loops also generate nonlocal terms such as curvature times $\log(-\Box/\mu^2)$ times curvature. Their coefficients depend on the light spectrum and observable.

The validity window requires small external momenta, curvature scales, and gradients relative to the cutoff, together with a controlled loop expansion. Near a strong-curvature region, an EFT prediction requires a new error estimate rather than formal continuation of a truncated series.

**Obstruction 11.4** (A derivative truncation is not an exact ultraviolet propagator)**.** *The exact toy inverse operator gives*

$$\frac{1}{x(1 + x/m^2)} = \frac{1}{x} - \frac{1}{x + m^2}, \tag{35}$$

*with opposite pole residues. Its low-energy expansion is instead*

$$\frac{1}{x} - \frac{1}{m^2} + \frac{x}{m^4} - \cdots. \tag{36}$$

*An extra pole inferred by exact resummation of a truncated EFT is not automatically a physical state within the EFT window.*

A fundamental nondegenerate higher-time-derivative theory and a perturbative EFT have different consistency questions. The spin-two ghost in the usual exact quadratic-gravity treatment is a genuine warning [21], but it does not exclude every curvature correction. For example, $R + \alpha R^2$ with $\alpha > 0$ and positive conformal factor has a healthy local scalar representation with $m_s^2 = 1/(6\alpha)$ in $M_{\mathrm{Pl}} = 1$ units. Degenerate constrained theories also require separate analysis.

## 12 Long-range exchange and quantum corrections

**Proposition 12.1** (Newton normalization)**.** *For the canonically normalized massless tensor with $\kappa^2 = 32\pi G$, the four-dimensional conserved-source numerator is*

$$P_{\mu\nu,\alpha\beta} = \tfrac{1}{2}(\eta_{\mu\alpha}\eta_{\nu\beta} + \eta_{\mu\beta}\eta_{\nu\alpha} - \eta_{\mu\nu}\eta_{\alpha\beta}). \tag{37}$$

*Two static sources give*

$$\widetilde{V}(\mathbf{q}) = -\frac{4\pi G m_A m_B}{\mathbf{q}^2}, \qquad V(r) = -\frac{G m_A m_B}{r}. \tag{38}$$

*Proof.* The numerator contraction is $m_A m_B/2$. The two $\kappa/2$ vertices give $-\kappa^2 m_A m_B/(8\mathbf{q}^2)$. Fourier transformation uses $1/\mathbf{q}^2 \mapsto 1/(4\pi r)$. Conserved sources remove longitudinal gauge terms. □

For minimally coupled massive sources in the conventional nonrelativistic scattering-potential matching prescription, the familiar long-range expression is [17]

$$V(r) = -\frac{G m_1 m_2}{r}\left[1 + 3\frac{G(m_1+m_2)}{rc^2} + \frac{41}{10\pi}\frac{G\hbar}{r^2c^3} + \cdots\right]. \tag{39}$$

The potential beyond leading order is a matching object, not a generally covariant local observable. Coordinate/canonical conventions and subtraction of iterated Born terms matter. The robust EFT content is the nonanalytic long-distance amplitude and its properly defined observable consequences. The quoted coefficient is not asserted for every matter spectrum or every potential convention.

With the stated convention the relative quantum term is $41\ell_P^2/(10\pi r^2)$, where $\ell_P^2 = G\hbar/c^3$. It is approximately $3.41 \times 10^{-70}$ at one meter and $1.26 \times 10^{-61}$ at $52\,\mu$m. These are calculated scales, not measured deviations. Local Wilson coefficients remain independent matching data even when a particular long-distance nonanalytic contribution is fixed.

## 13 Relational observables and gravitational constraints

Gauge-invariant predictions are functions of physical records, not coordinate labels. In canonical language, physical observables commute with first-class constraints on the physical state space, or are represented by an appropriate relational/dressed construction.

**Proposition 13.1** (Relational exchange example)**.** *For $C = Z \otimes I + I \otimes Z$, the operator $X \otimes X + Y \otimes Y$ commutes with $C$, while $X \otimes I$ does not.*

*Proof.* The exchange operator preserves total $Z$ charge; direct Pauli multiplication gives the stated commutators. □

This finite example demonstrates a constraint-invariant relational observable, not a solution of gravitational constraints. In the spatially compact vacuum setting, the local-observable obstruction excludes the relevant class of gauge-invariant observables built locally from finitely many metric derivatives [18]. It does not exclude relational, nonlocal, asymptotic, or matter-clock observables outside those hypotheses.

The classical hypersurface-deformation algebra contains, schematically,

$$\{H[N], H[M]\} = D[q^{ab}(N\partial_b M - M\partial_b N)], \tag{40}$$

together with the $D$–$D$ and $D$–$H$ brackets. The structure functions depend on the metric. A quantum representation must specify regulator, domains, operator ordering, boundary terms, physical states, and the continuum interpretation of closure. A small finite commutator residual is not enough.

**Proposition 13.2** (Finite truncation can hide an algebra defect)**.** *For the oscillator truncated to $N$ levels,*

$$[a_N, a_N^\dagger] = I - N|N-1\rangle\langle N-1|. \tag{41}$$

*The defect vanishes on the lower-level block but has norm $N$ on the full truncated space.*

*Proof.* The diagonal entries of the commutator are one except at the highest level, where the missing raising transition gives $-(N-1)$. Subtracting the identity gives the formula. □

The example is not a gravitational anomaly. It shows why cutoff-boundary and continuum estimates are required before declaring an operator algebra represented. Gauge constraints and gravitational dressing also obstruct naive factorization of the physical Hilbert space into independent spatial factors. Microscopic tensor composition and physical constrained factorization are different questions.

## 14 Finite-resource inference and ultraviolet nonuniqueness

An observational obstruction must distinguish exact equality of laws from small differences and must specify the permitted experiments and resource budget.

**Theorem 14.1** (Finite-resource discrimination bound)**.** *Suppose two models have single-trial laws $P, Q$ with $\mathrm{TV}(P,Q) \le \varepsilon$. For $N$ independent trials,*

$$\mathrm{TV}(P^{\otimes N}, Q^{\otimes N}) \le \min(1, N\varepsilon). \tag{42}$$

*For equal prior probabilities, achieving average error at most $\delta < 1/2$ requires*

$$N\varepsilon \ge 1 - 2\delta. \tag{43}$$

*The same bound holds for an adaptive classical transcript if the conditional next-record laws differ by at most $\varepsilon$ uniformly over all histories and allowed actions.*

*Proof.* A maximal coupling fails to match a single record with probability at most $\varepsilon$. Couple trials sequentially and use the union bound, giving mismatch probability at most $N\varepsilon$. For adaptive records the same argument applies while histories agree, using the uniform conditional bound. Equal-prior optimal error is $(1 - \mathrm{TV})/2$, which yields (43). □

For quantum channel uses, an analogous hybrid argument uses the half-diamond distance, uniformly over permitted ancillas and inputs. A bound only for one fixed measurement is insufficient to constrain arbitrary coherent adaptive experiments.

**Proposition 14.2** (A stronger regular statistical bound)**.** *If $D_{\mathrm{KL}}(P\|Q) \le \kappa$, then independent trials obey*

$$\mathrm{TV}(P^{\otimes N}, Q^{\otimes N}) \le \sqrt{N\kappa/2}. \tag{44}$$

*Consequently error at most $\delta$ requires $N \ge 2(1-2\delta)^2/\kappa$. In regular nearby families with $\kappa = O(\varepsilon^2)$, the necessary scaling is $\varepsilon^{-2}$ rather than the weaker $\varepsilon^{-1}$ bound.*

*Proof.* Relative entropy adds for independent products. Apply Pinsker's inequality and the equal-prior testing formula. For adaptive classical records the chain rule gives the same conclusion under a uniform conditional relative-entropy bound. □

**Corollary 14.3** (Resource-relative resolution horizon)**.** *If every permitted protocol has a uniform per-use law difference bounded by $\varepsilon$ and the total budget satisfies $N_{\max}\varepsilon < 1 - 2\delta$, no such protocol attains error $\delta$. This is an approximate finite-resource obstruction, not exact operational equivalence.*

An EFT suppression $(E/M)^n$ is not automatically a total-variation bound. Converting it into one requires an observable, coefficient bounds, response functions, noise, state preparation, and control of all allowed protocols. Rare events, resonances, accumulated phases, and coherent strategies can change sensitivity. A cosmic-mode count is not a universal count of independent identical trials. Therefore no table of nominal energies proves that every ultraviolet completion lies in the exact kernel of all accessible observations.

**Proposition 14.4** (Infrared data do not uniquely fix ultraviolet dynamics)**.** *If two models have the same retained low-energy coefficients and differ only in unmeasured higher-order terms, observations restricted to the retained order cannot select between them. Finite nonzero higher-order effects may nevertheless distinguish them with sufficient resources or a different protocol.*

*Proof.* At the retained order their predicted laws agree by construction. Beyond that truncation, equality has not been assumed, so no exact-kernel conclusion follows. □

Ultraviolet completion remains an open physical and mathematical problem. Neither a finite-resource bound nor a common Einstein limit proves that all candidate completions are consistent, equivalent, or permanently unobservable.

## 15 Further obstructions and their domains

**Composite gravitons.** The Weinberg–Witten result constrains massless spin greater than one carrying nonzero energy-momentum in a Lorentz-invariant setting with the relevant Lorentz-covariant conserved stress tensor [22]. A fixed-background composite-graviton proposal satisfying those hypotheses must address the obstruction. General relativity does not possess a local gauge-invariant gravitational stress tensor of the forbidden kind. Evading a hypothesis can remove the obstruction; it is not a construction of a consistent completion.

**Classical horizons.** For the fixed Schwarzschild geometry in ingoing Eddington–Finkelstein coordinates, outgoing radial null rays satisfy

$$\frac{\mathrm{d}r}{\mathrm{d}v} = \frac{1}{2}\left(1 - \frac{2M}{r}\right). \tag{45}$$

They do not escape from $r < 2M$ in that classical spacetime. This constrains direct causal access to interior events. It does not prove that all interior states have identical exterior quantum records: exterior fields, correlations, global constraints, and the quantum spacetime problem require separate analysis.

**Reversibility and reduced dynamics.** A reversible closed quantum evolution can induce irreversible reduced dynamics. A controlled-NOT acting on $|+\rangle|0\rangle$ produces an entangled pure state; discarding the second qubit leaves $I/2$. Thus Assumption A3 cannot be applied to every reduced subsystem to exclude decoherence. A measurement-and-feedback model may yield dephasing, but one such channel does not classify every classical–quantum hybrid. Decoherence–diffusion constraints are model-dependent physical tests [24], not a consequence of the polytope theorem alone.

**No unique ultraviolet theory from exclusions.** Eliminating polytopes under continuous transitivity, one matrix family under a composition rule, or an exact ghost pole does not exhaust all possible microscopic theories. Each exclusion is relative to a declared class. The derivation does not turn a finite list of no-go statements into an unrestricted uniqueness theorem.

## 16 A coherence-sensitive gravitational prediction

Assume the weak-field quantum Newton interaction

$$H_{\rm int} = -\frac{G m_A m_B}{|\hat{x}_A - \hat{x}_B|}. \tag{46}$$

For two equal-amplitude position branches per mass, negligible branch motion during the interaction, and controlled non-gravitational phases,

$$|\psi(t)\rangle = \frac{1}{2}\sum_{i,j=0}^{1} e^{i\varphi_{ij}}|ij\rangle, \qquad \varphi_{ij} = \frac{G m_A m_B t}{\hbar r_{ij}}. \tag{47}$$

**Theorem 16.1** (Entangling phase invariant)**.** *The phase*

$$\Theta = \varphi_{00} + \varphi_{11} - \varphi_{01} - \varphi_{10} \tag{48}$$

*is invariant under $\varphi_{ij} \mapsto \varphi_{ij} + \alpha_i + \beta_j$. The pure-state concurrence and negativity are*

$$\mathcal{C} = \left|\sin\frac{\Theta}{2}\right|, \qquad \mathcal{N} = \frac{1}{2}\left|\sin\frac{\Theta}{2}\right|. \tag{49}$$

*Proof.* The local phases cancel in $\Theta$. With coefficient matrix $A_{ij} = e^{i\varphi_{ij}}/2$, concurrence is $2|\det A| = |\sin(\Theta/2)|$. For a pure two-qubit state, the negative partial-transpose eigenvalue has magnitude equal to the product of Schmidt coefficients, which is half the concurrence. □

For $m_A = m_B = 10^{-14}\,\mathrm{kg}$, center separation $450\,\mu\mathrm{m}$, equal collinear branch splitting $250\,\mu\mathrm{m}$, and interaction time $2.5\,\mathrm{s}$, the stated geometry gives

$$\Theta = -0.313934492575, \qquad \mathcal{N} = 0.078161730662. \tag{50}$$

These are idealized calculated values in the assumed interaction model, with scales motivated by proposed entanglement experiments [23]. They are not observations. Finite wave-packet size, acceleration, trapping, environmental decoherence, electromagnetic interactions, and readout must be included in a physical error budget. Local dephasing can remove the ideal entanglement.

**Proposition 16.2** (Classical local-channel obstruction)**.** *A separability-preserving local-operation and classical-communication channel cannot entangle an initially separable pair of probes. In particular, a convex mixture of product unitaries cannot do so.*

*Proof.* Each product operation maps a product state to a product state, or a conditional product branch. Convex mixing and classical conditioning preserve the convex set of separable states. Iterating the operations preserves separability. □

Inferring a nonclassical mediator from entanglement additionally requires initially uncorrelated probes, only the stipulated local probe–mediator couplings, no unaccounted quantum communication, and a classical mediator model restricted to the relevant classical-information channel class. The calculation does not identify a unique ultraviolet mediator, nor exclude every nonlocal direct-interaction or hybrid model without those assumptions.

## 17 The derivation theorem and dependency ladder

**Theorem 17.1** (Conditional low-energy quantum gravity)**.** *Assume operational closure, the full quantum reconstruction conditions, an admissible four-dimensional causal continuum with calibrated volume, a covariant source realization, the healthy universal tensor sector and its nonlinear consistency conditions, and perturbative quantum consistency. Then:*

(i) *retained finite-dimensional records admit the complex quantum state/effect representation;*

(ii) *the causal realization and volume determine the physical Lorentzian metric within the admitted conformal class;*

(iii) *the source is the metric response, with its Ward identities, improvements, and renormalized correlations;*

(iv) *the leading two-derivative gravitational response is Einstein–Hilbert with universal coupling, and static tree exchange gives* $-Gm_Am_B/r$*;*

(v) *perturbative quantum predictions are organized by the gravitational EFT, not by Einstein-only closure;*

(vi) *physical predictions use gauge-invariant relational, dressed, asymptotic, or otherwise admissible observables;*

(vii) *the assumed coherent Newton interaction yields the invariant entangling phase of Theorem 16.1;*

(viii) *finite-resource discrimination is bounded by Theorem 14.1 and Proposition 14.2, without implying exact ultraviolet observational nullity.*

*Proof.* Apply Structural input 2.4 to the reconstruction assumptions. Causal reconstruction and Theorem 4.3 fix the metric within the admitted realization. Theorem 7.1 identifies the source response. Theorem 9.1, Structural input 9.3, and Corollary 9.4 give the leading tensor dynamics. Proposition 12.1 fixes the static normalization. Assumption A8, Structural input 11.2, and Theorem 11.3 give the quantum low-energy organization. The observable construction must respect the constraints described in Section 13. The last two conclusions are the separately proved phase and discrimination statements. □

The finite-range construction supplies an explicit compatible localization and free-continuum example. It is not a premise-free derivation of the four-dimensional continuum or the metric-response assumption. Conversely, the main theorem does not require that particular walk to be the microscopic physical theory.

| Step | Result | Dependence |
|---|---|---|
| 1 | Operational properties and predictive dynamics | Exact descent criterion; physical record family specified |
| 2 | Complex quantum representation | Full reconstruction assumptions and structural theorem |
| 3 | Coherence and phase invariants | Declared quantum realization and reference basis |
| 4 | Lorentzian metric | Admitted distinguishing continuum; causal reconstruction and calibrated volume |
| 5 | Exact finite propagation and free Dirac limit | Explicit one-dimensional finite-range unitary construction |
| 6 | Stress and quantum Ward identities | Covariant source functional and anomaly conditions |
| 7 | Einstein leading response | Linear selection proved; nonlinear consistency used structurally |
| 8 | Quantum gravitational EFT | Perturbative quantization, counterterms, matching, validity window |

| Step | Result | Dependence |
|---|---|---|
| 9 | Observable phase and discrimination | Declared interaction and experimental/resource assumptions |
| 10 | Nonperturbative completion | Not implied by the preceding steps |

## 18 Status ledger and remaining mathematical questions

The ledger records the precise scope and dependencies of the final statements.

| Component | Status | Exact scope or missing input |
|---|---|---|
| Operational descent | PROVED | Factorization criterion; microscopic closure must be demonstrated |
| Quantum realization | CONDITIONAL | Reconstruction axioms not derived from finite resolution alone |
| Composition identity | PROVED | All ranks within the declared same-type matrix family |
| Coherence geometry | PROVED LOCALLY | Fixed reference/support; positivity and stratum changes retained |
| Signature obstruction | PROVED | Positive real pullbacks cannot be Lorentzian |
| Order and volume | CONDITIONAL | Unique metric within admitted causal continuum and calibrated measure |
| Strict propagation | CONSTRUCTED | Explicit free one-dimensional walk; exact support bound |
| Free continuum limit | PROVED | Fixed time and momentum band; error bound (14) |
| Quantum balance | PROVED | Number-current operator identity; not yet gravitational stress |
| Metric/source identification | CONDITIONAL | Covariant functional and record correspondence assumed |
| Linear tensor response | PROVED | Declared local parity-even two-derivative ansatz |
| Nonlinear Einstein sector | CONDITIONAL | Structural consistency result with field-content hypotheses |
| Matter anomaly controls | CHECKED | Specified Standard-Model representation; not full quantum gravity |
| Perturbative EFT | CONDITIONAL | Gauge consistency, matching, and low-energy error control |
| Relational observables | REQUIRED | Full physical observable algebra not constructed here |
| Finite-resource inference | PROVED | Uniform law/channel bounds and explicit resource class |
| Gravitational entanglement | CONDITIONAL PREDICTION | Quantum Newton interaction and controlled protocol |
| Interacting $3+1$ limit | OPEN | Dynamics, stress limits, and continuum existence |
| Quantum constraints | OPEN | Regulated closure, domains, physical Hilbert space |
| Ultraviolet completion | OPEN | Existence, unitarity, singular regimes, physical selection |

*Open problem* 1 (Microscopic selection). Derive or empirically select the state-space, composition, and causal-realization assumptions from a concrete microscopic record/update law. A finite list of excluded alternatives is insufficient.

*Open problem* 2 (Interacting continuum and stress convergence). Construct an interacting four-dimensional limit with uniform estimates controlling correlation functions, renormalized source variations, causal propagation, and any defect stress. The free fixed-band theorem does not establish this limit.

*Open problem* 3 (Physical constraint representation). Construct a regulated quantum representation of the gravitational gauge/constraint identities with controlled domains and continuum limit, a physical state space, and a sufficient algebra of relational or dressed observables.

*Open problem* 4 (Ultraviolet existence and selection). Establish a consistent completion, its controlled infrared matching, and predictions distinguishing it from other completions with the same retained EFT coefficients. Finite-resource limitations do not close this question.

# 19 Conclusion

The derivation identifies a precise low-energy endpoint: a quantum state/effect theory, a causally and volumetrically calibrated Lorentzian realization, a covariant stress response, a universal Einstein leading tensor sector, and its perturbative gravitational EFT. The construction includes exact descent criteria, an all-dimension composition identity, finite propagation, an operator continuity law, a controlled free continuum bound, the linear tensor selection, and quantitative inference and entanglement formulas.

The remaining assumptions are physical and mathematical content, not merely notation. In particular, a local net is not yet a dynamical metric, internal phase curvature is not spacetime curvature, a finite resource horizon is not exact observational nullity, and a perturbative EFT is not a nonperturbative ultraviolet completion. The low-energy construction is consequently predictive within its stated window while leaving nonperturbative existence and microscopic selection as distinct mathematical questions.

# A Computational verification and reproducibility

The accompanying calculations use constructed mathematical states and exact symbolic expressions; they are not experimental data. Finite-model and curvature verification consists of eighteen executed groups without a failed assertion. Their numerical outputs are used only for the explicitly stated constructions and illustrative values. The interpretation of a theorem is determined by its hypotheses and proof, not by a passing assertion or a descriptive string in a script.

The supplementary verification executes sixteen checks, all passing with seed 20261110. The source and machine-readable output are

`rg_quantum_gravity_derivation_checks.py`
`rg_quantum_gravity_derivation_checks.json`.

The checks include:

1. the symbolic all-dimension composition factorization;
2. the massive dispersion expansion and analytic group-velocity canary;
3. the finite-time free Dirac bound on a fixed momentum band;
4. the discrete current as an operator identity;

5. graded CAR locality versus even-observable commutativity;
6. calibrated Poisson volume and its statistical error;
7. exact product-law discrimination with nonzero total variation;
8. global reversibility with reduced dephasing;
9. the additional color and weak-doublet anomaly controls;
10. conserved improvements and unequal fluctuations at equal mean source;
11. phase-cycle invariance and the symbolic linear gauge identity;
12. local dephasing of the ideal entanglement prediction;
13. the finite-truncation boundary defect.

Dependencies are Python, NumPy, SciPy, and SymPy. Source hashes and result values are recorded in the machine-readable outputs. Determinism is understood within the stated software environment; byte-identical floating-point output across arbitrary library versions is not assumed. Numerical tests do not prove the imported reconstruction, nonlinear completion, two-loop calculation, or quantum constraint closure.

## B Normalization and approximation conventions

The spacetime signature is $(-,+,+,+)$ and $c = \hbar = 1$ except in dimensional estimates and the entanglement example. The reduced Planck mass satisfies $M_{\mathrm{Pl}}^{-2} = 8\pi G$ and $\kappa^2 = 32\pi G$. The mass parameter $M$ in the free Dirac limit is distinct from a generic EFT cutoff; its meaning is local to Section 5. The Schwarzschild parameter is in geometrized units. The SLD/Bures normalization is $\mathrm{d}s_B^2 = F_Q \mathrm{d}\theta^2/4$. Total variation is half the $L^1$ distance, and relative entropy uses natural logarithms. Testing errors use equal priors unless stated otherwise.

The bibliography identifies the structural literature used in the derivation. Empirical scales are illustrative inputs rather than claims of the latest experimental limits. No new literature verification or fit to experimental data is asserted.

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